a converging mirror with positive power. Also, a silvered lens is basically a modified mirror. Thus,

$$P = \frac{1}{-f};\ P_l = \frac{1}{f_l};\ P_m = \frac{1}{-f_m} \qquad (6.79)$$

Now equation (6.78) becomes,

$$\left(\frac{1}{-f}\right) = \left(\frac{2}{f_l}\right) + \left(\frac{1}{-f_m}\right) \qquad (6.80)$$

Proper sign conventions are to be followed for equation (6.80).

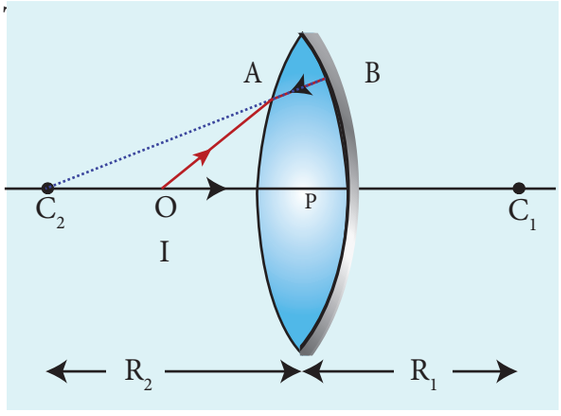
Suppose the object distance u and image distance v are to be found, we can very well use the mirror equation (6.8), since the silvered lens is a modified mirror.

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

EXAMPLE 6.18

A thin biconvex lens is made up of a glass of refractive index 1.5. The two surfaces have equal radii of curvature of 30 cm each. One of its surfaces is made reflecting by silvering it from outside. (a) What is the focal length and power of this silvered lens? (b) Where should an object be placed in front of this lens so that the image is formed on the object itself?

Solution



Given, n = 1.5; R_1 = 30 cm; R_2 = –30 cm;

(a) Let us find f_l and f_m separately.

Using lens maker's formula we can find f_l.

$$\frac{1}{f_l} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

Substituting the values, $\frac{1}{f_l} = (1.5-1)\left(\frac{1}{30} - \frac{1}{-30}\right)$

$$\frac{1}{f_l} = (0.5)\left(\frac{1}{30} + \frac{1}{30}\right) = \left(\frac{1}{2}\right)\left(\frac{2}{30}\right) = \frac{1}{30\,\text{cm}}$$

f_l = 30 cm = 0.3 m

Focal length of mirror is, $f_m = \frac{R_2}{2}$

Substituting the values, $f_m = \frac{-30}{2} = -15\,\text{cm}$

f_m = 15 cm = –0.15 m

Now the focal length of the slivered lens is,

$$\frac{1}{-f} = \frac{2}{f_l} + \frac{1}{-f_m} = \frac{2}{30} + \frac{1}{15} = \frac{2}{15} = \frac{1}{7.5}$$

f = – 7.5 cm = –0.075 m

The silvered mirror behaves as a concave mirror with its focal length on left side.

To find the power of the silvered lens, $P = 2P_l + P_m$

$$P = \frac{2}{f_l} + \frac{1}{-f_m} = \frac{2}{0.3} + \frac{1}{-(-0.15)}$$

$$= \frac{2}{0.3} + \frac{1}{0.15} = \frac{4}{0.3} = 13.33\,D$$

As the power is positive it is a converging system.

[Note: Here, we come across a silvered lens which has negative focal length and positive power. Which implies that the focal length is to the left and the system is a converging one. Such situations are possible in silvered lenses because a silvered lens is basically a modified mirror.]

(b) Writing the mirror formula,

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

Here, both u and v are same ($v = u$) as the image coincides with the object.

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